

WS06: Trajectory of a Small Ball

Time: 1 hr

Name: ()

Class: **26S0**
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Date:

This investigation considers the trajectory of a small ball as it rolls off a surface which is inclined to the horizontal.

- (a) (i)** Set up the apparatus as shown in Fig 1.

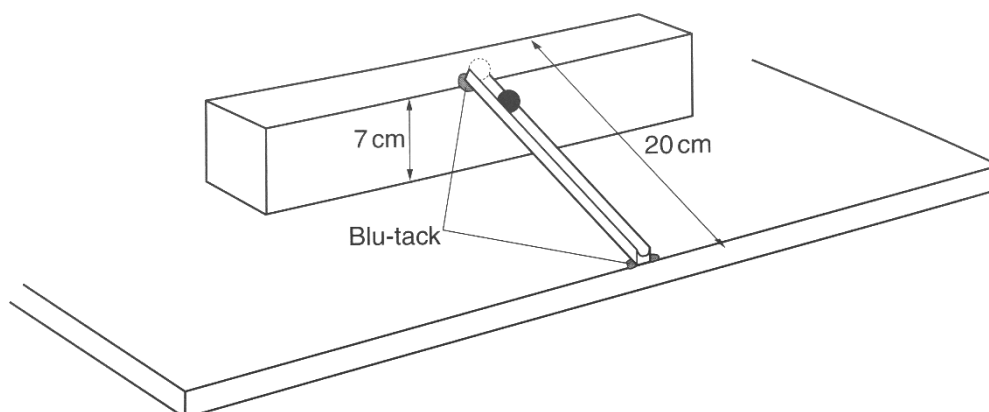


Fig. 1

- (ii)** Tie a string to the pendulum bob to be used as a plumbline.
- (iii)** Suspend the plumbline from the edge of the bench using adhesive tape, with the plumbline aligned with the bottom of the ramp, as shown in Fig 1.

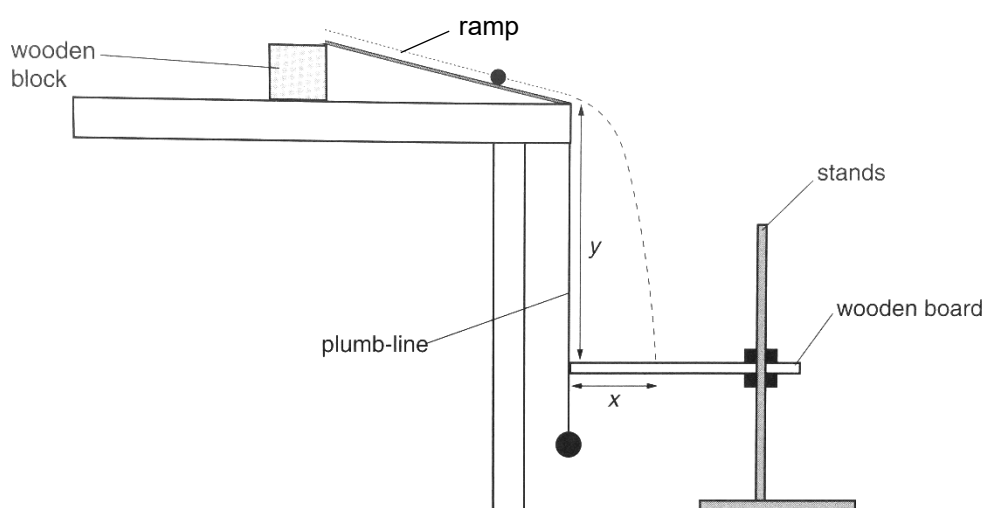


Fig. 2

- (iv)** Mount a wooden board horizontally using two clamps mounted on two retort stands so that the board is situated about 60 cm below the bottom of the ramp.
- (v)** Use a pencil to make a small mark on the top of the board to mark the contact point between the edge of the board and the plumbline as shown in Fig. 1.3.

- (vi) Attach a blank piece of paper onto the board using adhesive tape such that the short edge of the paper is aligned with the edge of the board furthest from the mark, as shown in Fig. 1.3.

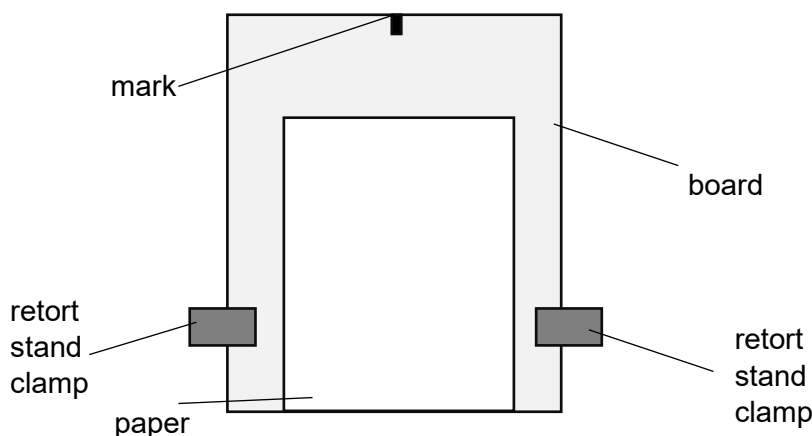


Fig. 3 (top view of board)

- (vii) Place a sheet of carbon paper on top of the blank paper. The ink-covered side of the carbon paper should be facing down.
- (b) (i) Position the ball on top of the ramp. Release the ball so that it rolls down the ramp and onto the board below. Lift up the carbon paper slightly and observe that the ball makes an imprint on the blank paper.
- (ii) Measure and record the vertical distance y (between the bottom of the ramp and the top of the board) and the horizontal distance x (between the mark and the imprint).

Vertical distance between bottom of ramp and top of board

$$y = 0.598 \text{ m},$$

Horizontal distance of marble

$$x_1 = 0.280 \text{ m}, \quad x_2 = 0.273 \text{ m}$$

$$\langle x \rangle = \frac{0.280 + 0.273}{2} = 0.277 \text{ m}$$

$$y = 0.598 \text{ m}$$

$$x = 0.277 \text{ m} \quad [2]$$

- (iii) Estimate the percentage uncertainty in the value of x in (ii).

$$\Delta x = \frac{1}{2} |x_{\max} - x_{\min}| = \frac{1}{2} (0.280 - 0.273) = 0.0035 \approx 0.004 \text{ m}$$

Percentage uncertainty

$$\frac{\Delta x}{x} \times 100\% = \frac{0.004}{0.277} \times 100\% = 1.4\% \quad (2 \text{ s.f.})$$

$$\text{percentage uncertainty} = 1.4\% \quad [1]$$

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- (c) Reduce the value of y and repeat steps (b)(i) and (b)(ii) to obtain further sets of readings for x and y . Present your results clearly.

y / m	x_1 / m	x_2 / m	$\langle x \rangle / \text{m}$	$\frac{y}{\langle x \rangle}$
0.589	0.280	0.273	0.277	2.13
0.484	0.248	0.248	0.248	1.95
0.384	0.217	0.215	0.216	1.78
0.311	0.203	0.202	0.203	1.54
0.236	0.164	0.162	0.163	1.45
0.137	0.117	0.120	0.119	1.16

[5]

- (d) Theory suggests that x and y are related by the equation

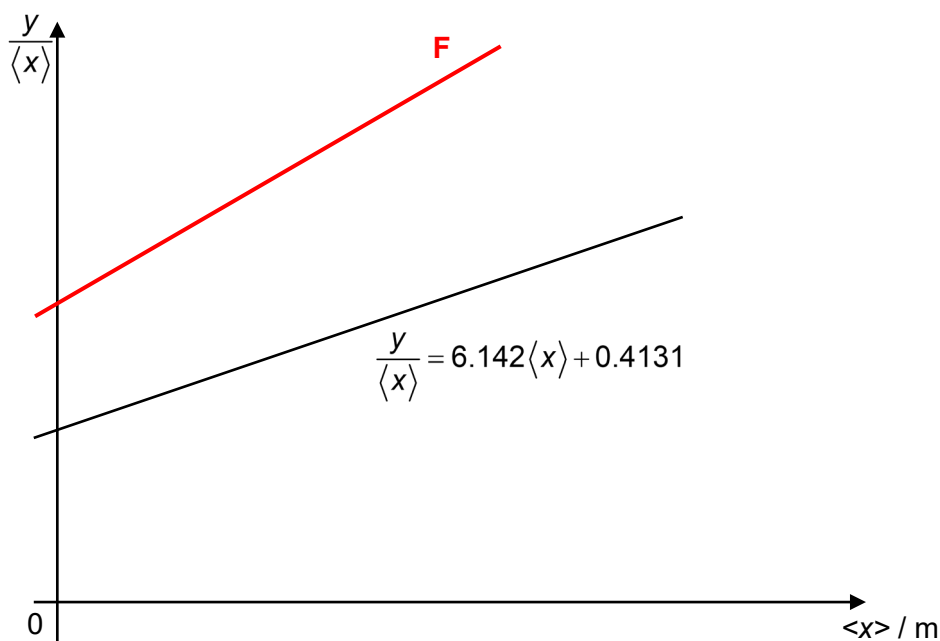
$$\frac{y}{x} = \frac{g(1+k^2)x}{2u^2} + k$$

where $g = 9.81 \text{ m s}^{-2}$, u is the speed of the ball as it leaves the ramp and k is a constant.

Plot a suitable graph using a spreadsheet to determine the values of u and k .

Sketch the graph in the space below. Write down the equation of the trendline.

Plot a graph of $\frac{y}{\langle x \rangle}$ against $\langle x \rangle$, where $\frac{g(1+k^2)}{2u^2}$ is the gradient and k is the vertical intercept.



$k = \text{vertical intercept} = 0.4131$

$$\frac{g(1+k^2)}{2u^2} = \text{gradient} = 6.142$$

$$\frac{(9.81)(1+0.4131^2)}{2u^2} = 6.142$$

$$u = 0.967 \text{ m s}^{-1}$$

$$u = 0.967 \text{ m s}^{-1}$$

$$k = 0.4131 \quad [2]$$

- (e) The angle of inclination θ of the ramp is related to k by the equation:

$$k = \tan \theta$$

- (i) Calculate the angle of inclination θ_c of the ramp.

$$\tan \theta = 0.4131$$

$$\theta = \tan^{-1}(0.4131) = 22.4^\circ$$

$$\theta_c = 22.4^\circ \quad [1]$$

- (ii) Take measurements to determine the angle of inclination θ_m of the ramp.

$$\theta_1 = 20^\circ$$

$$\theta_2 = 20^\circ$$

$$\theta = \frac{\theta_1 + \theta_2}{2} = 20^\circ$$

$$\theta_m = 20^\circ \quad [1]$$

- (iii) Calculate the percentage difference between these two values.

$$\text{percentage difference} = \frac{22.4^\circ - 20^\circ}{20^\circ} \times 100\% = 12\%$$

$$\text{percentage difference} = 12\% \quad [1]$$

- (f) The experiment is repeated by increasing the angle of inclination of the ramp and releasing the ball from the same vertical height above the bench.

On the graph in (d), sketch a second graph to represent the new results. Label the line F.

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- (g) (i) State one significant source of error in this experiment.
1. The marble may be released from different positions at the top of the ramp. This results in an inconsistent value of x .
 2. Sideways motion as the marble leaves the track causes greater uncertainty in x .
 3. The carbon mark made by the marble is quite large and the precision in the measurement of x is affected.
- (ii) Suggest an improvement that could be made to the experiment to address the source of error identified in (g)(i). You may suggest the use of other apparatus or a different procedure.
1. Make a mark on the ramp as a reference position and release the marble from the same position on the ramp each time.
 2. Use a track with a V-shaped cross section to minimise sideways Use a track with a V-shaped cross section to minimise sideways
 3. Use a smaller and denser marble (or steel ball) for a more precise point of contact with the wooden board.

END OF WORKSHEET